Sahaj Bhadja

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Summary

Software engineer with 2+ years building testable cloud applications, REST APIs, microservices, data pipelines, and AI-assisted workflows using Python, Java, TypeScript, React, SQL, AWS, Azure, Docker, CI/CD, and Agile practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant **Azure SQL and Blob Storage** services for 500+ users, optimizing schemas, indexed queries, and secure access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **FastAPI workflows** for 1,000+ monthly AI jobs using explicit states, retries, validation, and failure isolation, tripling throughput while improving production reliability and debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Python REST APIs and React workflows** handling 10,000+ daily requests, integrating secure document ingestion, AWS RDS persistence, and AI-assisted processing while maintaining stable production performance.
- Designed concurrent batch processing for 1,000+ files using multithreading, automated validation, and fault-tolerant execution, reducing turnaround by 60% while strengthening testing and defect resolution under peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible Python data pipelines processing 100,000+ trajectory logs, improving long-horizon planning by 18% through systematic testing, debugging, machine-learning evaluation, and documented experiment workflows.
- Architected a spatial-temporal encoder combining HD-map geometry and historical trajectories through multi-head attention, improving closed-loop simulation stability by 15% across complex, high-volume analytical data workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **microservices** with Jenkins CI/CD and 20+ PyTest and Mockito suites, automating unit, integration, and regression testing across Windows, Linux, and macOS production environments.
- Optimized **REST API and ETL integrations** using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a **React and Node.js** platform with REST APIs, WebSockets, and eight modular services, supporting 2M+ synchronized records while translating requirements into maintainable real-time business workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production releases.

Skills

Languages & Applications: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, React, Node.js, FastAPI, Flask, REST APIs, JSON, WebSockets, microservices, object-oriented programming, design patterns, data structures, algorithms; **Data & AI:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Firebase, AWS RDS, ETL, data integration, data pipelines, analytics, machine learning, NLP, LLM orchestration, semantic search, AI-assisted development, human-in-the-loop workflows; **Cloud & Quality:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, code review, performance optimization, monitoring, Agile, Scrum, SDLC, technical documentation, security, compliance workflows

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a scalable financial platform supporting 50,000+ users and 1M+ transactions with secure Plaid integration, real-time synchronization, advanced filtering, and containerized deployment across distributed services.

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI knowledge platform for 200+ users with secure ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ governed documents.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, REST APIs*
- Built an AI-assisted Java and TypeScript database workspace supporting 10+ systems, accelerating SQL authoring, schema exploration, validation, and relational data management through a unified cross-platform interface.